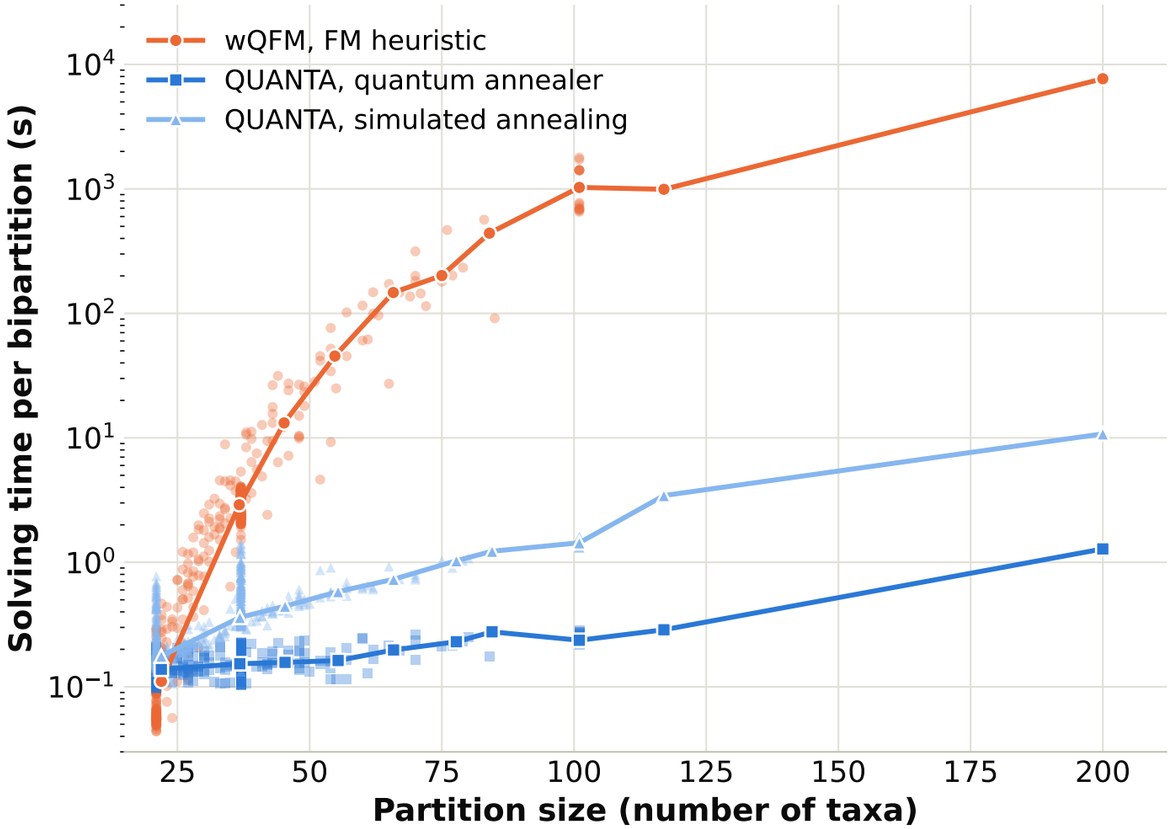
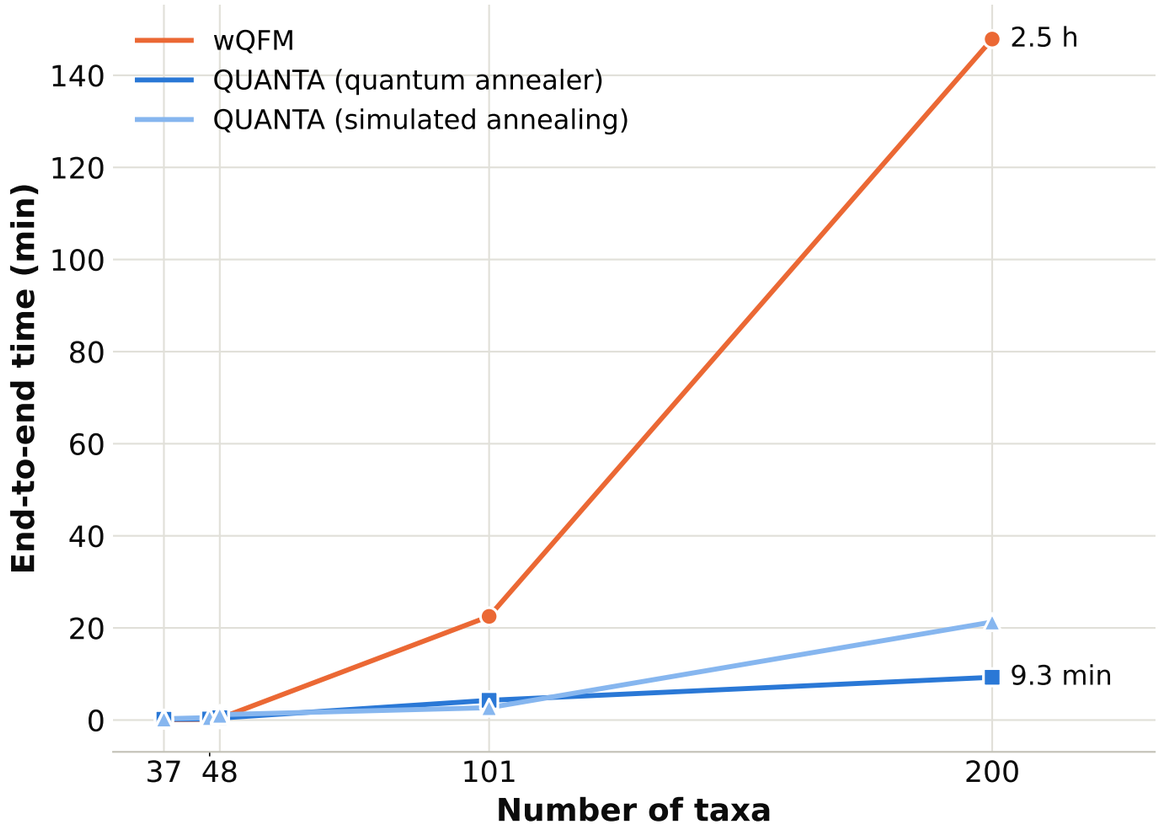
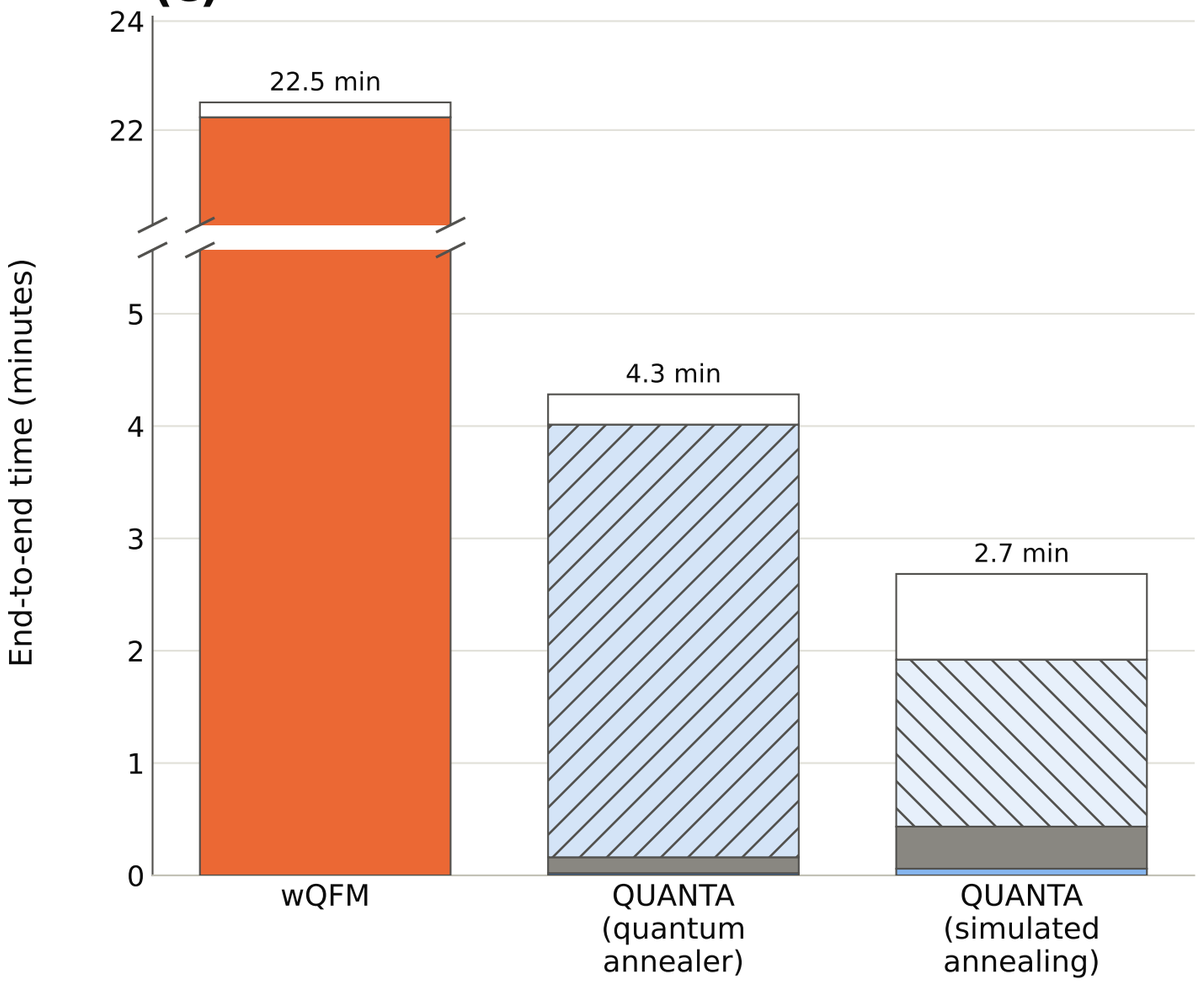
(a)



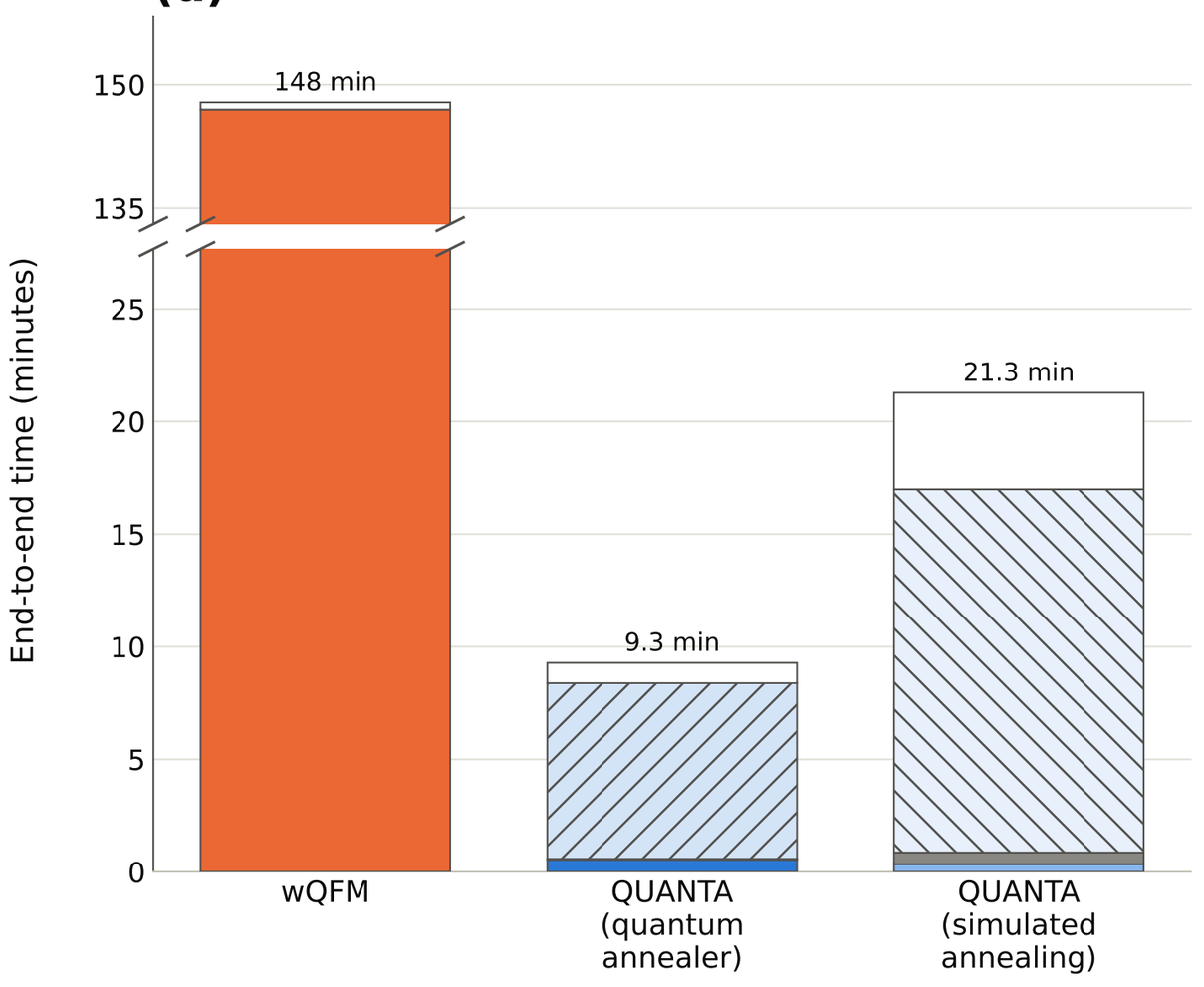
(b)



(c)



(d)



wQFM: core FM step
QUANTA: quantum computer time
QUANTA: simulated annealing time
Exact search (partitions ≤ 20 taxa)
QUANTA (annealer): QUBO building + D-Wave access
QUANTA (sim. annealing): QUBO building
Rest of the run (reading quartets, merging, etc.)